

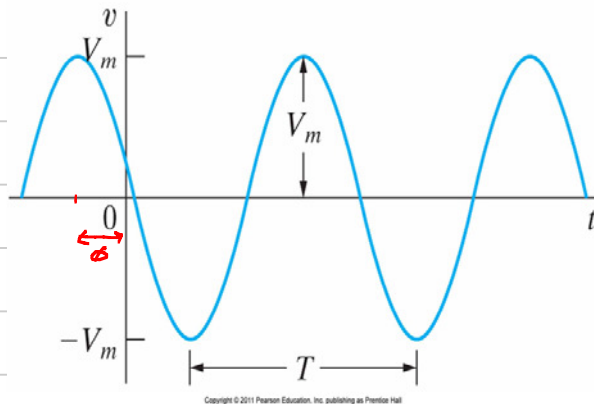


Chapter 5, Sinusoidal Steady State Response

In this chapter we will analyze circuits that are energized by time varying voltage or current sources. In particular we are interested in sinusoidal sources. A sinusoidal signal is easy to generate and transmit. The form of voltage generated at power plants and supplied to homes and factories is the sinusoidal voltage. A sinusoidal source is usually referred to as alternating current or ac source.

A sinusoidal voltage is expressed as:

$$V(t) = V_m \cos(\omega t + \phi) \quad \text{where,} \quad \begin{aligned} V_m &= \text{amplitude (V)} \\ \omega &= \text{angular freq } \left(\frac{\text{rad}}{\text{s}}\right) \\ &\quad \left(\omega = 2\pi f = \frac{2\pi}{T}\right) \\ \phi &= \text{phase angle} \end{aligned}$$



Either sine or cosine function can be used to express a sinusoidal signal.

We will use the cosine function throughout our discussion.

To convert sine to cosine function subtract 90° from the phase angle.

If the amplitude is negative add or subtract 180° and make it positive.

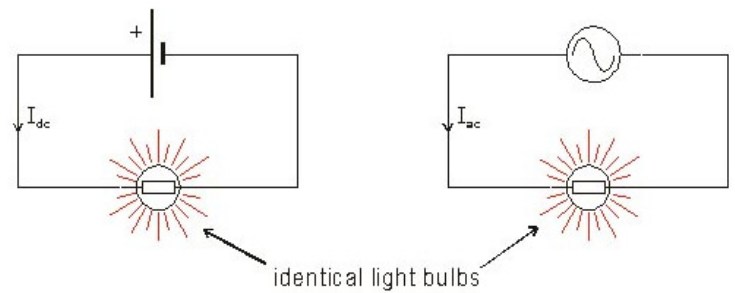
$$\sin \omega t = \cos(\omega t - 90^\circ)$$

$$-\cos(\omega t) = \cos(\omega t \mp 180^\circ)$$

rms value:

The rms value of an ac voltage is its equivalent dc value that delivers the same amount of power to two identical loads.

If $P_1 = P_2$, then the effective value of I_{ac} is I_{dc} .



The rms value of a periodic function is defined as the square root of the average of the squared function.

$$V_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} V(t)^2 dt}$$

For a sinusoidal voltage :

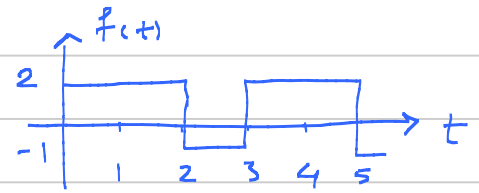
$$V_{rms} = \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi) dt}, \text{ using } \cos^2 \theta = \frac{1 + \cos 2\theta}{2} :$$

$$V_{rms} = V_m \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} \frac{1 + \cos 2(\omega t + \phi)}{2} dt}$$

average of a sinusoid = 0

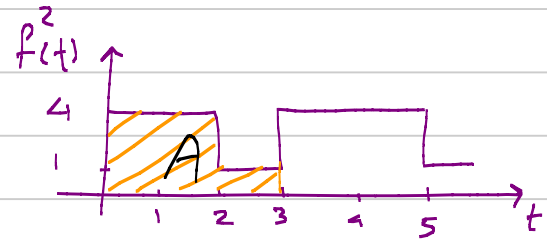
$$= \frac{V_m}{\sqrt{2}} \sqrt{\frac{1}{T} (T) + \frac{1}{T} \int_{t_0}^{t_0+T} \cos 2(\omega t + \phi) dt} \Rightarrow \boxed{V_{rms} = \frac{V_m}{\sqrt{2}}}$$

Ex. Find the rms value of $f(t)$.



$$f_{\text{rms}} = \sqrt{\frac{1}{3} \int_0^3 f^2(t) dt} = \sqrt{\frac{1}{3} (A)}$$

$$= \sqrt{\frac{1}{3} (8+1)} = \sqrt{3}$$



Ex. $v(t) = 10 \cos(377t - 53.13^\circ)$. Find a) f , b) T , c) V_m , d) ϕ .

a) $\omega = 2\pi f = 377 \rightarrow f = 60 \text{ Hz}$

b) $T = \frac{1}{f} = 0.0167 \text{ ms}$,

c) $V_m = 10 \text{ V}$,

d) $\phi = -53.13^\circ$

$$= -53.13 \times \frac{\pi}{180} = -0.93 \text{ rad}$$

Ex. Find the phase angle between i_1 and i_2 . Does i_1 lead or lag i_2 ?

$$i_1 = -4 \sin(377t + 25^\circ) \quad , \quad i_2 = 5 \cos(377t - 40^\circ)$$

First convert i_1 to the standard form (cosine with a positive amplitude)

$$\begin{aligned} i_1 &= -4 \cos(377t + 25^\circ - 90^\circ) = -4 \cos(377t - 65^\circ) \\ &= 4 \cos(377t - 65^\circ + 180^\circ) = 4 \cos(377t + 115^\circ) \end{aligned}$$

Phase difference : $115 - (-40) = 155^\circ$ (i_1 leads i_2 by 155°)

Phasors

The response of an ac circuit is the sum of the transient response and the steady state. The transient response is dictated by the nature of the circuit. The steady state has the same form as the excitation. If a circuit is excited by a sinusoidal source, the steady state response is also sinusoidal at the same frequency as the source but different amplitude and phase angle.

Because of this characteristic, we don't have to carry the frequency in our calculation. **Phasor is a complex number that carries the amplitude and phase angle of a sinusoidal function.**

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi \quad (\text{Euler's identity})$$

$$\cos \phi = \operatorname{Re}\{e^{j\phi}\} \quad , \quad \sin \phi = \operatorname{Im}\{e^{j\phi}\}$$

$$\begin{aligned} V &= V_m \cos(\omega t + \phi) = V_m \operatorname{Re}\{e^{j(\omega t + \phi)}\} = V_m \operatorname{Re}\{e^{j\omega t} e^{j\phi}\} \\ &= \operatorname{Re}\{V_m e^{j\phi} e^{j\omega t}\} \end{aligned}$$

$V_m e^{j\phi}$ is a complex number that carries the amplitude and

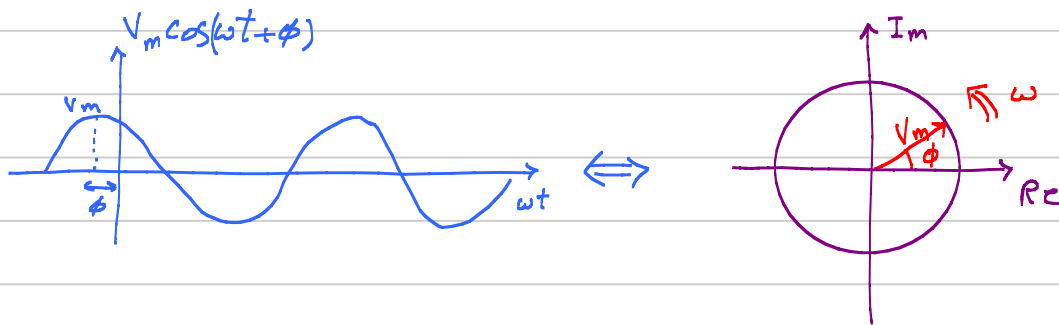
phase angle of a sinusoidal function. This complex number is

by definition the phasor transform of a given sinusoidal function.

time domain

phasor domain

$$\begin{aligned} V = V_m \cos(\omega t + \phi) &\Leftrightarrow V = V_m e^{j\phi} && \text{(exponential form)} \\ &= V_m \cos \phi + j V_m \sin \phi && \text{(rectangular form)} \\ &= V_m \angle \phi && \text{(polar form)} \end{aligned}$$



Ex. Express these sinusoids as phasors;

$$V = -7 \cos(2t + 40^\circ), \quad i = 4 \sin(10t + 10^\circ)$$

$$V = 7 \cos(2t + 40^\circ - 180^\circ) = 7 \cos(2t - 140^\circ)$$

$$\Rightarrow V = \underline{7 \angle -140^\circ}$$

$$i = 4 \sin(10t + 10^\circ) = 4 \cos(10t + 10^\circ - 90^\circ) = 4 \cos(10t - 80^\circ)$$

$$\Rightarrow I = 4 \angle -80^\circ$$

Complex calculations (review)

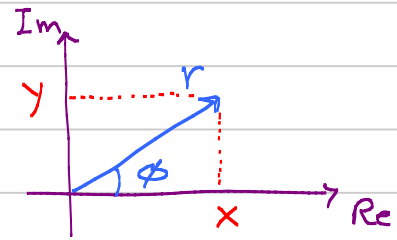
$Z = x + jy$ (rectangular form) , x : real part, y : imaginary part

$Z = r \angle \phi$ (polar) , r : amplitude , ϕ : phase angle

$Z = r e^{j\phi}$ (exponential form) $\rightarrow Z = r (\cos \phi + j \sin \phi)$

Polar to rectangular: $x = r \cos \phi$, $y = r \sin \phi$

rectangular to polar: $r = \sqrt{x^2 + y^2}$, $\phi = \tan^{-1} \frac{y}{x}$



If $Z_1 = x_1 + jy_1 = r_1 \angle \phi_1$, $Z_2 = x_2 + jy_2 = r_2 \angle \phi_2$

Then: $Z_1 + Z_2 = (x_1 + x_2) + j(y_1 + y_2)$

$Z_1 - Z_2 = (x_1 - x_2) + j(y_1 - y_2)$

$Z_1 Z_2 = r_1 r_2 \angle \phi_1 + \phi_2$

$\frac{Z_1}{Z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$

$\frac{1}{Z_1} = \frac{1}{r_1} \angle -\phi_1$

Complex conjugate of Z_1 is:

$Z_1^* = x_1 - jy_1 = r_1 \angle -\phi_1 = r_1 e^{-j\phi_1}$

Phasor transform of circuit elements

If we find the V-I relationship of the circuit elements, in phasor domain, we can find the phasor transform of the element.

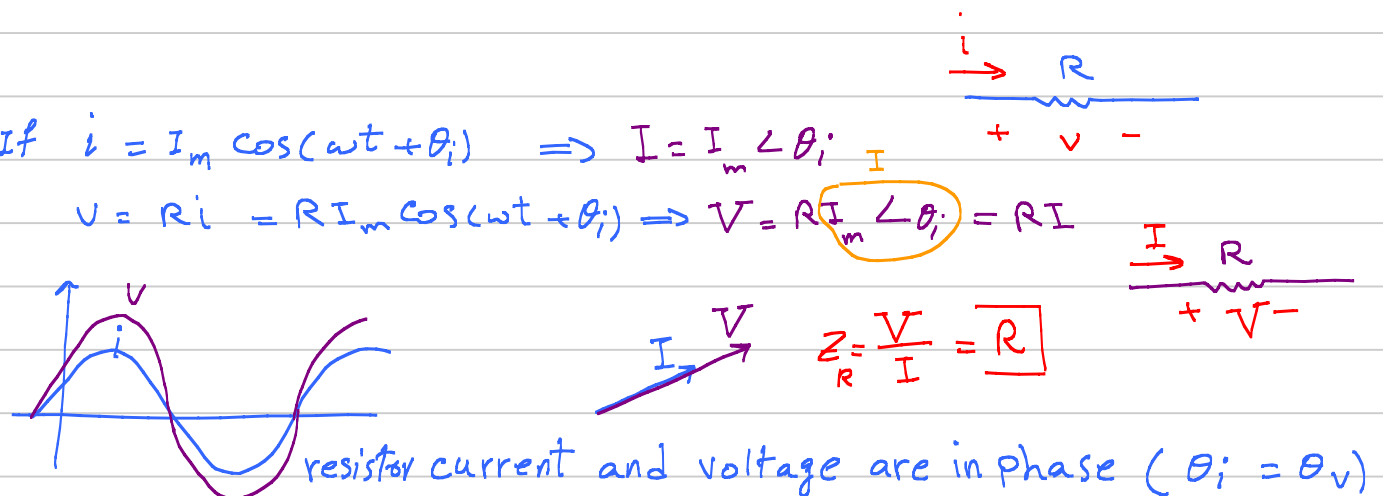
Resistor in phasor domain

If $i = I_m \cos(\omega t + \theta_i) \Rightarrow I = I_m \angle \theta_i$

$V = Ri = RI_m \cos(\omega t + \theta_i) \Rightarrow V = RI_m \angle \theta_i = RI$

$Z_R = \frac{V}{I} = R$

resistor current and voltage are in phase ($\theta_i = \theta_v$)



Inductor in phasor domain

$i = I_m \cos(\omega t + \theta_i) \Rightarrow I = I_m \angle \theta_i$

$V = L \frac{di}{dt} = LI_m [-\omega \sin(\omega t + \theta_i)] = \omega LI_m \cos(\omega t + \theta_i + 90^\circ)$

$\Rightarrow V = \omega LI_m e^{j(\theta_i + 90^\circ)} = \omega LI_m e^{j\theta_i} e^{j90^\circ} \leftarrow e^{j90^\circ} = \cos 90^\circ + j \sin 90^\circ = j$

$V = j\omega L I \Rightarrow Z_L = \frac{V}{I} = j\omega L$

Inductor current lags behind the voltage by 90° .

